Pranshav Patel

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Education

North Carolina State University Aug 2025 – May 2027
Master of Computer Science *GPA: 4.0/4.0*
- Coursework: Software Engineering, DevOps, Design and Analysis of Algorithms, Adv. Topics in ML and Neural Networks

Nirma University Sep 2021 – Jun 2025
Bachelor of Technology in Computer Science *GPA: 8.58/10.0*
- Coursework: Database Management Systems, Data Structures and Algorithms, Software Engineering, Machine Learning

Work Experience

Data Science Intern, *LexisNexis Legal & Professional* *Jun 2026 – Aug 2026*
- Re-architected an **agentic LLM pipeline** for legal due diligence (**Python, Google ADK, LiteLLM, FastAPI, Pydantic**), replacing a nondeterministic LLM planning step with a deterministic, **budget-capped evidence-retrieval layer** driven by versioned JSON review plans.
- Cut **LLM prompt tokens by 66%, inference cost by 63%, and end-to-end latency by 38%** (mean across 20 controlled A/B experiments) while keeping output quality on par with the production baseline.
- Designed a reproducible **A/B evaluation harness** with frozen inputs, isolated per-arm processes, input hashing, automated quality gates, and **LLM-as-judge** scoring to measure every change against the production baseline.
- Shipped the layer as a **feature-flagged**, opt-in capability across 3 runtime modes, with per-document fallback ensuring backward compatibility with production behavior.

Software Engineer Intern, *Trading Technologies Pvt. Ltd.* *Dec 2024 – Jun 2025*
- Automated more than **80%** of manual regression test cases for the SCORE compliance product, reducing execution time from **6 hours to under 1 hour**.
- Developed Python scripts to validate critical financial data transformations (KRM to RISK, WFE pipelines) and automated **RESTful API** validation, reducing manual verification effort by **70%**.
- Built scalable automation frameworks on **AWS** (S3, DynamoDB, EC2, EMR, Glue Jobs), accelerating data ingestion and validation pipelines by **35%**.
- Partnered cross-functionally with QA and engineering teams to standardize test automation practices, reducing release-cycle bottlenecks by **40%**.

Projects

PolicyBot: Full-Stack AI Assistant — React, JavaScript, Python, FastAPI, LangGraph
- Built and deployed a **full-stack AI assistant end-to-end** (**React** front end, **FastAPI** backend) live on Vercel/Render, with a single **JWT** role-based access and audit layer shared by the REST API, chat agent, and a **Slack** bot.
- Built hybrid **BM25** + dense retrieval with cross-encoder reranking (**Recall@1 92.5%**, **MRR 0.958** on 67 labeled questions); traced the last miss to a chunking bug that stripped section headers, fixed it, and added a groundedness check that blocks unsupported numbers in answers.
- Built interchangeable hand-rolled **ReAct** and **LangGraph** agent engines with tool calling and per-user conversation memory; fixed a production bug in both that wrote duplicate leave requests by making successful tool calls terminal.

Nanoserve: Single-GPU LLM Inference Engine — Python, PyTorch, CUDA, FlashInfer, Hugging Face, vLLM
- Built an **end-to-end Qwen2.5 inference engine** with custom model execution, SafeTensors loading, **paged KV caching, continuous batching, recompute preemption, and an OpenAI-style HTTP/SSE streaming API.**
- Integrated an opt-in FlashInfer paged-decode backend that increased throughput by **81% (75.4 → 136.6 output tokens/sec), reduced p99 time-to-first-token from 19.2 s to 145 ms**, and **eliminated 92/387 request failures** on an NVIDIA L40S; validated against Hugging Face and a same-session vLLM baseline over 3,691 replayed requests.

Fitbit-MCP: Personal Health Data Platform — Python, MCP, SQLite
- Built a local-first **MCP server with 13 tools** for Fitbit and Google Health data, with OAuth authentication and resumable imports into **SQLite**; reduced dashboard rendering from **45 seconds to 3 seconds** across **2 million heart-rate samples** through indexing and caching.
- Developed an **end-to-end agent evaluation harness** spanning **32 synthetic workflows and 192 model trials**, checking tool calls, database state, exports, and multi-turn authorization; identified confirmation-expiry failures and added automated regression tests.

Skills

Languages: Python, JavaScript, SQL, C/C++, Java
Frontend & Backend: React, Tailwind CSS, Node.js, FastAPI, Pydantic, REST APIs, SSE Streaming, JWT, RBAC
Databases: MySQL, SQLite, MongoDB, Redis, DynamoDB
AI/LLM: LLM Agents, Tool Calling, RAG, LangGraph, Google ADK, LiteLLM, MCP, LLM-as-Judge Evals
Performance: PyTorch, Hugging Face, vLLM, FlashInfer, KV Caching, Continuous Batching, Benchmarking
Cloud & DevOps: AWS (S3, SQS/SNS, EC2, EMR, Glue), Docker, CI/CD